INTERNATIONAL SEARCH REPORT

International application No.
PCT/KR2020/003809

A. CLASSIFICATION OF SUBJECT MATTER

F21K 9/66(2016.01)i, H01L 33/54(2010.01)i, H01L 33/10(2010.01)i, F21K 9/64(2016.01)i, F21K 9/68(2016.01)i

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

F21K 9/66; F21Q 1/00; F21S 2/00; F21S 8/10; F21V 5/04; H01L 33/00; H01L 33/54; H01L 33/10; F21K 9/64; F21K 9/68

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched
Korean utility models and applications for utility models: IPC as above
Japanese utility models and applications for utility models: IPC as above

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

eKOMPASS (KIPO internal) & Keywords: board, light, resin, reflect, convex, curve

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	CN 101876407 A (FOSHAN INSTITUTE OF SUN YAT-SEN UNIVERSITY et al.) 03 November 2010 See paragraphs [0022]-[0025] and figure 1.	1-13
A	JP 2007-158009 A (NICHIA CHEM IND LTD.) 21 June 2007 See paragraphs [0050]-[0054] and figure 10.	1-13
A	CN 101868865 B (KONINKL PHILIPS ELECTRONICS NV.) 22 August 2012 See paragraphs [0036]-[0038] and figure 1.	1-13
A	KR 10-2016-0047249 A (LG INNOTEK CO., LTD.) 02 May 2016 See claims 1-2 and figure 1a.	1-13
A	JP 10-321017 A (ICHIKOH IND LTD.) 04 December 1998 See claim 1 and figure 4.	1-13
A	KR 10-2015-0138710 A (LG INNOTEK CO., LTD.) 10 December 2015 See paragraphs [0094]-[0138] and figure 15.	1-13

☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
06 JULY 2020 (06.07.2020)	**06 JULY 2020 (06.07.2020)**
Name and mailing address of the ISA/KR Korean Intellectual Property Office Government Complex Daejeon Building 4, 189, Cheongsa-ro, Seo-gu, Daejeon, 35208, Republic of Korea Facsimile No. +82-42-481-8578	Authorized officer Telephone No.

Form PCT/ISA/210 (second sheet) (January 2015)

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/KR2020/003809

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